Computer Organization
Homework 2
(Due on 2018/10/25 8:00PM)

Note: Please hand in the homework on time.

1. For the following C statement, what is the corresponding MIPS assembly code? Assume that the variables x, y, and z are saved in $s0, $s1, $s2. Please include instruction "addi". (10%)

$x = (y + 8) - (z - 5);$

2. For the following MIPS assembly instructions above, what is a corresponding C statement? (10%)

$sub \quad f, \quad c, \quad d;$
$sll \quad e, \quad f, \quad 4;$

3. Assume that the variables i and j are assigned to register $s2, $s3. Assume that the base address of arrays P and Q are in register $s6 and $s7, respectively. Variable ACC are assigned to register $s5. For the MIPS assembly instruction below, what is the corresponding C statement? Please answer (a) to (e) step by step. (20%)

sll	$t0, $s2, 2	
add	$t0, $s6, $t0	
sw	$s5, 0($t0)	# (a) What is the meaning of these three lines (Please use the variables in the statement of the question if the registers are defined.)
sll	$t2, $s3, 2	
add	$t2, $s7, $t2	
lw	$t0, 4($t2)	# (b) What is the meaning of these three lines
add	$s5, $s5, $t0	# (c) What is the meaning of this line?
lw	$t0, -4($t2)	# (d) What is the meaning of this line?
sub	$s5, $s5, $t0	# (e) What is the meaning of this line?

4. Translate the following C code to MIPS. Assume that the variables i is assigned to register $s0. Assume that the base address of arrays A and B are in register $s4 and $s5, respectively. (20%)

$$B[i] = A[2i] - A[2i - 1]$$

5. Assume that registers $s0 and $s1 hold the values 0x6FFFFFFF and 0x90000002, respectively.

(a) What is the value of $t7 for the following assembly code? Please express in hexadecimal. (5%)

```
sub $t7, $s0, $s1
```

(b) What is the result in $s2 and $s3 for the following instruction? (10%)

```
slti $s2, $s1, -1
sltu $s3, $s0, $s1
```

6. Assume that the base address of arrays A and B are in register $s4 and $s5 as shown in the following figures.

(a) What are the values in $s0 and $s3? (10%)

(b) What is the value in $t2 after executing the following instructions? (5%)

```
        sub    $s3, $s6, $s7
        beq    $s3, $s1, Thisway
        add    $t2, $s2, $s7
        sw     $t2, 12($s4)
        lw     $s0, 16($s4)
        j      Exit
Thisway: addi  $t2, $s2, -49
        sw     $t2, 4($s5)
        lw     $s0, 0($s5)
Exit:   ....
```

(c) After executing all the instructions, which element in array A/B is updated? And what is the new value saved in the memory? (10%)

Register

$s0	$s1	$s2	$s3	$s4	$s5	Ss6	$s7
??	10	98	??			16	6

Memory

5172	B [1]=27
5168	B [0]=9
5164	
5160	
5156	A [4]=16
5152	A [3]=8
5148	A [2]=4
5144	
5140	